MID-SEMESTER RETEST EXAMINIATION, APRIL 2025

MAT 2226 – ENGINEERING MATHEMATICS – IV

Date:17-04-2025 **Time: 5:00-6:30 PM** **Marks: 25**

Q. No.	Description	Marks
11.	A bag contains 4 red balls, 5 blue balls, and 6 green balls. Two balls are drawn one after another without replacement. Define the events: A: The first ball drawn is red. B: The second ball drawn is blue. C: At least one of the two balls drawn is green. a. Compute $P(A \cup B)$. b. Given that at least one ball is green, what is the probability that the first ball was red?	4
12.	Suppose that the accident rate in a certain state is four accidents per one million inhabitants per month. a. Find the probability that in a certain town with a population of 5, 00, 000 there will be at most 4 accidents in a month. b. What is the probability that during one year there are at least two months in which more than 4 accidents occur?	3
13.	If (X, Y) is a continuous two-dimensional random variables having joint pdf $f(x,y) = \begin{cases} e^{-(x+y)}; & x \geq 0, y \geq 0 \\ 0; & otherwise \end{cases}$ then show that X and Y are independent. Also, evaluate $i)\ P(X < Y)\ ii) P(X + Y > 1)$.	3
14.	A fair coin is tossed thrice. Let X = 0 or 1 according to head or tail occurs on the first toss and Y = number of heads. Determine the joint probability distribution of X and Y. Also obtain V(X) and V(Y).	3
15.	Suppose that the 2-dimensional random variables (X, Y) has joint pdf $f(x,y) = \begin{cases} \frac{2}{\pi}; & x^2 + y^2 \leq 1, \quad y \geq 0 \\ 0; & Otherwise \end{cases}$. Evaluate correlation coefficient between X and Y..	3
16.	The random variable X has pdf $f(x) = \frac{3}{4}(2x - x^2),\ 0 \leq x \leq 2$. a. Show that $f(x)$ is a valid pdf of X. b. Find the cdf of X . c. Find a such that $P(X < a) = 2P(X > a)$.	3
17.	Suppose there is a specified list of spam words and that you have a database of 5000 emails of which 1700 are spam. Among the spam emails, 1343 emails contain words in the list. Of the 3300 non-spam emails, only 297 emails contain words in the list. What is the probability that an email contains words in the list?	2
18.	A town has two fire engines operating independently. The probability that a specific engine is available when needed is 0.96. a. what is the probability that neither is available? b. What is the probability that the fire engine is available when needed?	2
19.	If $A_1, A_2, \ldots, A_n$ are events from the sample space S, then prove that $P(\bigcap_{i=1}^{n} A_i) \geq \sum_{i=1}^{n} P(A_i) - (n-1)$.	2